Research Project - Mathematical Statistics

Ahmet Batuhan Uluçay

3247069
January 2025

1.Introduction

For this research project, I aim to predict an album's popularity based on its musical attributes as provided by Spotify's API service. Spotify measures an album's success and the songs in it through a popularity level based on number of streams, user interactions like saves and shares, and the track's inclusion in popular playlists. This study seeks to determine whether specific attributes can influence an album's popularity using the dataset. To ensure the feasibility of this study, I will operate under the following assumptions: The popularity measure of an album in Spotify also reflects the music preferences of the world as a whole (reasonable considering the power of the platform in the music industry). External factors, such as the artist's existing fame or marketing efforts, are excluded from consideration. Lastly, all songs within an album are assumed to be released simultaneously. This excludes the potential impact of releasing singles prior to an album's release, as this practice can significantly affect the album's popularity. The study will be focusing on the 2010s, due to the amount of elements within the dataset, its relative recency, and its uniqueness in being able to witness the rise of online music streaming platforms and their effect on the industry and the consumers preferences.

2.Data Cleaning

The dataset used in this study contains basic information about each album, most importantly its popularity score. Additionally, it includes various musical attributes describing three songs from each album. I made several preprocessing and data cleaning actions to fit the dataset to the study.I started by removing attributes that are unnecessary for the purposes of this study. Albums with missing values were also eliminated, as the dataset contains enough data to justify removing incomplete entries. I then converted the release dates of the albums to **Date** format and extracted their release years and added a decade column to group them accordingly.

To facilitate analysis at the album level, I generated new columns to calculate the average values of the musical attributes of the three songs from each album. These averages will be used to determine the musical characteristics of the album as a whole. Next, I created interaction terms (e.g., **dance_energy, dance_valence, energy_valence**) to examine the potential combined effects of closely related musical attributes on an album's popularity. To finish, I split the decade in half. A schematic description of such features, with the abbreviations used in the R script, follows:

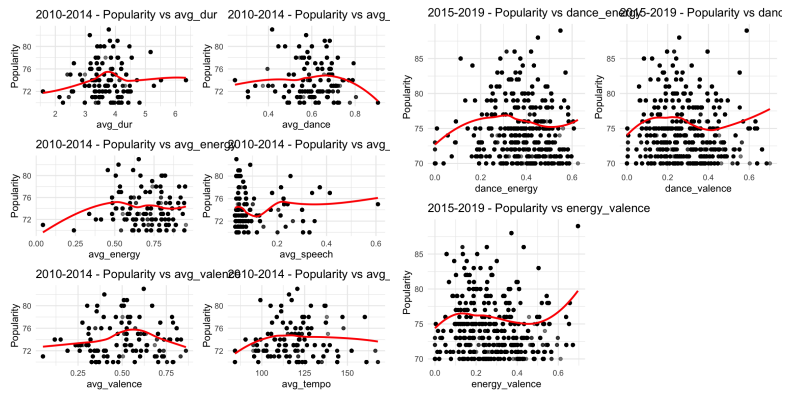
id: A unique identifier for each album in the dataset.
name: The name of the album.
popularity: A numerical score indicating the album's popularity, as determined by Spotify's algorithm.
release_year: The year the album was released.
decade: A categorical variable grouping albums into their respective decades based on release year.
avg_dur: The average duration (in minutes) of the three sampled songs from the album.

avg_dance: The average danceability score of the three songs, indicating how suitable they are for dancing.
avg_energy: The average energy score of the three songs, reflecting their intensity and activity.
avg_speech: The average speechiness score of the three songs, representing the presence of spoken words.
avg_valence: The average valence score of the three songs, which measures their musical positivity or mood.
avg_tempo: The average tempo (in beats per minute) of the three songs.
dance_energy, energy_valence, dance_valence: Interaction terms

3.Plotting and formation of Hypothesis

I made linear plots with every music attribute against popularity for both halves of the decade. I observed that most attributes had a weak or flat linear relationship with popularity. However, noticeable clustering in specific regions of the plots suggested the potential presence of non-linear relationships. Using **Loess**, I observed that most attributes demonstrated non-linear relationships with popularity. For example, **avg_dance** and **avg_valence** showed popularity peaking at moderate levels rather than steadily increasing or decreasing. Interaction terms like **dance_energy** and **dance_valence** have significant non-linear relationships with popularity, especially in the second half of the decade, resulting with the following hypothesis' that I will try to deduce in this study:

1. "It is possible to predict and determine album popularity using only the musical attributes we got, without considering external factors like an artist's fame or marketing."
2. "Spotify had an effect on listener preferences that can be seen by comparing the first and second half of the decade."



4.Regression models and diagnostics

I applied both linear and non-linear regression models to analyze the relationships between musical attributes and album popularity. I created separate models for the first half (2010–2014) and the second half (2015–2019) of the decade. Linear regression models initially revealed weak or flat relationships between most musical attributes and popularity with low R^2 for both halves. The **LOESS** regression models demonstrated significant non-linear relationships, revealing that attributes such as **avg_dance**, **avg_val, dance_energy dance_valence** had popularity peaks at moderate levels rather than linear increases or decreases. These findings were particularly pronounced in the second half of the decade.

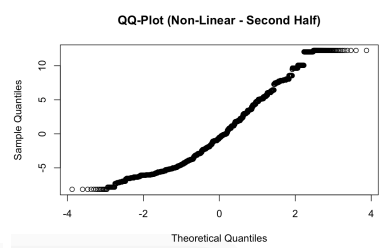
However, both linear and non linear regressions demonstrated low R^2 values, suggesting that musical attributes alone cannot fully explain popularity, supporting the idea that external factors.

Several diagnostics were conducted to evaluate model performance. **QQ-plots** of residuals showed deviations from normality for both linear and non-linear models, with more severe deviations observed in linear models. **The Kolmogorov-Smirnov** normality test confirmed that residuals were not normally distributed for all models, with **p-values** well below 0.05. Additionally, the **Breusch-Pagan** test identified heteroskedasticity in the residuals across all models, indicating that the variance of residuals was not constant. **Residual analysis** further supported these findings, as scatterplots of residuals against fitted values exhibited non-random patterns for the linear models. In contrast, residuals from the non-linear models were more aligned with the data.

The models and their diagnostics support the initial findings from the scatterplots; there seems to be a much stronger non-linear relation between the attributes and popularity levels than linear ones. However, low R^2 scores for both of them implies that musical attributes alone cannot fully explain popularity, supporting the idea that external factors. Furthermore, Differences in coefficient signs and significance between the two periods indicate changes in the relationships between attributes and popularity.

```
Call:
lm(formula = popularity ~ avg_dur + avg_dance + avg_energy +
    avg_speech + avg_valence + avg_tempo + dance_energy + dance_valence +
    energy_valence, data = first_half)

Residuals:
    Min      1Q  Median      3Q     Max
-6.5084 -2.7058 -0.4845  2.3034  7.4694

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     64.951864   1.307271  49.685  < 2e-16 ***
avg_dur          0.737489   0.097654   7.552 5.86e-14 ***
avg_dance        6.531359   3.324959   1.964 0.049595 *
avg_energy       6.946925   2.354922   2.950 0.003206 **
avg_speech       2.488264   0.659630   3.772 0.000165 ***
avg_valence     14.849813   3.347440   4.436 9.53e-06 ***
avg_tempo        0.003763   0.004399   0.855 0.392418
dance_energy   -11.114818   5.635836  -1.972 0.048694 *
dance_valence   -3.620395   3.451406  -1.049 0.294291
energy_valence -12.183446   3.583104  -3.400 0.000683 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.29 on 2644 degrees of freedom
Multiple R-squared:  0.07084,   Adjusted R-squared:  0.06768
F-statistic:  22.4 on 9 and 2644 DF,  p-value: < 2.2e-16
```

```
Call:
lm(formula = popularity ~ poly(avg_dance, 2) + poly(avg_energy,
    2) + poly(avg_valence, 2) + dance_energy + dance_valence +
    energy_valence, data = first_half)

Residuals:
   Min     1Q Median     3Q    Max
-5.195 -2.727 -0.386  2.351  8.961

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)            62.015      2.414  25.694  < 2e-16 ***
poly(avg_dance, 2)1   -69.335     22.699  -3.055  0.00228 **
poly(avg_dance, 2)2   -31.626      6.381  -4.956 7.63e-07 ***
poly(avg_energy, 2)1   24.250     21.524   1.127  0.26001
poly(avg_energy, 2)2   -29.374      5.162  -5.690 1.41e-08 ***
poly(avg_valence, 2)1 -354.136     49.349  -7.176 9.27e-13 ***
poly(avg_valence, 2)2  -75.716      6.300 -12.018  < 2e-16 ***
dance_energy           -24.937      5.741  -4.343 1.46e-05 ***
dance_valence           56.569      6.399   8.840  < 2e-16 ***
energy_valence          16.380      4.890   3.350  0.00082 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.241 on 2644 degrees of freedom
Multiple R-squared:  0.09829,   Adjusted R-squared:  0.09523
F-statistic: 32.02 on 9 and 2644 DF,  p-value: < 2.2e-16
```



5. ANOVA

I ran the ANOVA test two times. For the first hypothesis, I compared the linear and non linear models of the two half-decades. Both halves had a decreased RSS, indicating that non-linear is a better fit. F-statistics measure the improvement when going from the linear to non linear models. This was especially high for the second half. Finally, p-values for both half-decades are extremely small **Pr(>F),** once again showing that nonlinearity adds value to the model
Despite the statistically significant improvement, the overall R^2 values from the models remain low. For the first hypothesis, this suggests that while musical attributes are predictive, they are not the sole determinants of popularity.

```
Model 1: popularity ~ avg_dur + avg_dance + avg_energy + avg_speech +
    avg_valence + avg_tempo + dance_energy + dance_valence +
    energy_valence
Model 2: popularity ~ poly(avg_dance, 2) + poly(avg_energy, 2) + poly(avg_valence,
    2) + dance_energy + dance_valence + energy_valence
  Res.Df   RSS Df Sum of Sq F Pr(>F)
1   2644 28614
2   2644 27769  0    845.35
```

```
Model 1: popularity ~ avg_dur + avg_dance + avg_energy + avg_speech +
    avg_valence + avg_tempo + dance_energy + dance_valence +
    energy_valence
Model 2: popularity ~ poly(avg_dance, 2) + poly(avg_energy, 2) + poly(avg_valence,
    2) + dance_energy + dance_valence + energy_valence
  Res.Df    RSS Df Sum of Sq F Pr(>F)
1   9434 174850
2   9434 171324  0    3525.9
```

For the second hypothesis, I made the **half_decade** column a factor so that it treats the two periods as categorical. I then created two models, with one where **half_decade** interacts with the musical attributes and one where it doesn't to see if the relations differ in two decades. The residual sum of squares decreases when interaction terms are added, meaning Model 2 fits the data better. F-statistics found again indicates that the interaction terms improve the model fit. We also once again see a very small p-value.

The small p-value shows that the relationships between musical attributes (e.g., **avg_dance, avg_energy**) and popularity changed significantly between the first and second half of the decade.

```
Model 1: popularity ~ avg_dur + avg_dance + avg_energy + avg_speech +
    avg_valence + avg_tempo + half_decade
Model 2: popularity ~ (avg_dur + avg_dance + avg_energy + avg_speech +
    avg_valence + avg_tempo) * half_decade
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1  12090 209531
2  12084 208505  6    1026.2 9.9126 6.129e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

6.Conclusion

The research explored two hypotheses: first, whether album popularity could be predicted using only musical attributes without accounting for external factors like artist fame or marketing, and second, whether Spotify influenced listener preferences between the first and second halves of the decade. Regression models, both linear and nonlinear, were applied to test these hypotheses. The analysis revealed that musical attributes such as danceability, energy, and valence significantly influenced popularity, with non-linear models demonstrating superior fit compared to linear models, as indicated by ANOVA results. However, the low adjusted R^2 values across all models (below 10%) suggest that musical attributes explain only a small portion of the variance in popularity, highlighting the critical role of external factors. For the second hypothesis, interaction terms highlighted significant differences in how musical attributes related to popularity across the two periods. ANOVA tests showed a statistically significant improvement when accounting for these interactions, supporting the idea that Spotify, with their algorithms and user personalization, may have influenced shifts in music consumption trends. I considered including LASSO in this study. However, due to the fact that it cannot directly handle nonlinear terms, and the fact that it became clear early in the study that nonlinear methods were more significant, I decided against it.

The limitations of this study are clear from the assumptions made in the introduction. An extra attribute that was a popularity level measuring that of the artist like that of an album would potentially result in a more deciding study, and could have been used by parties of interest in the music industry. This would allow us to not largely offset the inaccuracy caused by the assumption that external factors didn’t play a role in an albums popularity.

A similar study with individual songs instead of albums can also enhance the results of this study and also allow us to find new conclusions. As I mentioned in another assumption, the rise of online streaming platforms heavily favoured the release of songs as singles rather than with an album. This new study would potentially allow us to determine if albums had lost in importance over time, further enhancing our knowledge.

Bibliografia

- Link to Dataset: https://www.kaggle.com/datasets/elemento/music-albums-popularity-prediction
- Wikipedia
- An Introduction to Mathematical Statistics, Bijma et al.
- https://www.geeksforgeeks.org/non-linear-regression-in-r/
- https://www.geeksforgeeks.org/residual-analysis/